Which of the following reason(s) lead towards the numerical integration methods?

Select the correct option

- All above choices are true

Correct

- Integrand is given in tabular form
- Analytical evaluation of integral is impossible
- Analytical evaluation of integral is very complicated

Geometrically the definite integral of any continuous function f(x) in the interval [a,b] gives --------.

Select the correct option

- ○ Volume with dimensions f(x), 'a' and 'b'
- ● Area under f(x) on [a,b]
- ○ Area of Trapezoid with dimension of 'a' and 'b'

correct

- ○ Length of segment AB on real line

While employing Trapezoidal and Simpson Rules to evaluate the double integral numerically, by using Trapezoidal and Simpson rule over ——.

Select the correct option

- Plane region
- Real line

$yn+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ is known as:

Select the correct option

○	Runge-Kutta's formula correct
○	Taylor's formula
○	Simpson's formula
○	Euler's formula

In double integration, we keep one variable say x fixed and .....

Select the correct option

- Reliable the other variable y
- Varying the other variable y

correct

Trapezoidal rule of integration of a definite integral is of.....

Select the correct option

- None of the given choices
- O(h4)
- O(h2) correct
- O(h3)

Which of the following method is simplest one to integrate numerically a given tabular function but give more error?

Select the correct option

- Simpson's 1/3 Rule
- Trapezoidal method
- Rectangular method
- Simpson's 3/8 Rule

Correct

The predictor-corrector method an _____ method.

Select the correct option

- direct
- implicite
- explicit
- indirect

correct

Given that $\frac{dy}{dt} = 1 - \sqrt{y}$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 3rd term of Taylor series when t=0.5 and y// =0.25.

Select the correct option

Re

○	0.01425
○	0.01325
○	0.01625
○	0.01525

The area of a trapezoid is obtained by adding the area of a .....and a triangle.

Select the correct option

- ○ Circle
- ○ Square
- ○ Rectangle
- ○ None of the given choices

correct

Multistep method does not improves the accuracy of the answer at each step.

Select the correct option

- FALSE
- TRUE

correct

At which of the following points the Maximum value of 2nd derivative of function
$f(x) = -(2/x)$ in the interval:[1,4] exits?

Select the correct option

- At x=1
- At x=4
- At x=3
- At x=2

correct

At which of the following points the Minimum value of 2nd derivative of function
f(x) = -(2/x) in the interval:[1,4] exits?

Select the correct option

- At x=1
- At x=4
- At x=2
- At x=3

correct

In Simpson's rule, we can estimate the integral by .....the areas under the parabolic arcs through three successive points.

Select the correct option

○	None of the given choices
○	Adding
○	Subtracting
○	Multiplying

correct

Adams – Bashforth is a multistep method.

Select the correct option

- FALSE
- TRUE

correct

Simpson's rule is a numerical method that approximates the value of a definite integral by using .....polynomials.

Select the correct option

- Quadratic
- Cubic
- None of the given choices
- Linear

correct

Trapezoidal and Simpson's integrations are just a linear combination of values of the
given function at different values of the ·····variable.

Select the correct option.

○	Arbitrary
○	Dependent
○	Independent
○	None of the given choices

correct

At which of the following points the Maximum value of 2nd derivative of function
$f(x) = -(2/x)$ in the interval:[1,4] exits?

Select the correct option

- At x=2
- At x=1
- At x=3
- At x=4

correct

Generally, Adams methods are superior if output at _____ points is needed.

Select the correct option

- many
- single
- at most
- two

correct

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 2nd term in Taylor series when t=1, y/ =1.2, and h=0.1.

Select the correct option

Reload

- 1.3
- 1.4
- 1.2
- 1.5

correct

In multistep method, if we choose small step size which _____ the accuracy level.

Select the correct option

○	increase
○	decrease

correct

In solving the differential equation

$$y' = x + y \; ; \; y(0) = 1$$

$$h = 0.2$$

By Euler's method y(0.2) is calculated as

$$y_1 = y_0 + hf(t_0, y_0),$$

Select the correct option

○	1.02
○	0.2
○	1.01
○	0.1

=1+0.2(0+1)
=1+0.2
=1.2

correct

Multistep method does not improves the accuracy of the answer at each step.

Select the correct option

- TRUE
- FALSE

correct

Question # 9 of 10 (Start time: 06:09:59 PM, 06 March 2022)

If $yn+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_3 is:

Select the correct option

○	$K_3 = hf(t_n - \frac{h}{2}, y_n - \frac{k_2}{2})$
○	$K_3 = hf(t_n + \frac{h}{2}, y_n + \frac{k_2}{2})$
○	$K_3 = hf(t_n, y_n)$
○	$K_3 = hf(t_n + \frac{h}{2}, y_n + \frac{k_2}{2})$

correct

In Trapezoidal rule, we assume that f(x) is continuous on [a, b] and we divide [a, b] into n subintervals of equal length using the .....points.

Select the correct option

- n
- n+1
- None of the given choices
- n-1

correct

If $\frac{dy}{dx} = \frac{y+x}{y-x}, y_0 = 1 \; at \; x_0 = 0$ and h= 0.2. Then, K_1 of Runge-Kutta method is:

Select the correct option

○	0.4
○	0.1
○	0.3
○	0.2

correct

To evaluate a definite integral of tabular function f(x), piecewise linear approximation led to --------.

Select the correct option

- Romberg's Method
- Simpson's 1/3 Rule
- Trapezoidal Method
- Simpson's 3/8 Rule

correct

To evaluate numerically a double integral over a rectangular region bounded by the lines $x = a$, $x = b$, $y = c$, $y = d$ we shall employ either trapezoidal rule or Simpson's rule, repeatedly with respect to .....variable at a time.

Select the correct option

- One
- Three
- Two
- None of the given choices

correct

If $yn+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_1 is:

Select the correct option

○	$K_1 = hf(t_n, yn+1)$
○	$K_1 = hf(t_n, y_n)$
○	$K_1 = hf(tn+1, y_n)$
○	$K_1 = hf(t_0, y_0)$

correct

Generally, Adams methods are superior if output at many points is needed.

Select the correct option

- FALSE
- TRUE

correct

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 1$ and $h = 0.1$. Then, K_1 of Runge-Kutta method is:

Select the correct option

- 0.4
- 0.3
- 0.1
- 0.2

hf(t0,y0)

0.1(1+1)

0.1(2)

0.2

Correct

In solving the differential equation

$$y' = x + y \; ; \; y(0) = 1$$

$$h = 0.2$$

By Euler's method y(0.2) is calculated as

Select the correct option

$$y_1 = y_0 + hf(t_0, y_0),$$

$$y(0.2) = 1 + (0.2)(0+1)$$

$$= 1.02$$

○	1.02
○	1.01
○	0.1
○	0.2

While employing Trapezoidal and Simpson Rules to evaluate the double integral numerically, by using Trapezoidal and Simpson rule with respect to -------- variable/variables at time

Select the correct option

- both
- single

correct

If $yn+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_2 is:

Select the correct option

- $K_2 = hf(t_n + \frac{h}{2}, y_n + \frac{k}{2})$
- $K_2 = hf(t_n + \frac{h}{2}, y_n + \frac{k_1}{2})$
- $K_2 = hf(t_n - \frac{h}{2}, y_n - \frac{k_1}{2})$
- $K_2 = hf(t_n, y_n)$

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 1$ find the 2nd term in Taylor series when t=1.5 and y/ =1.2

Select the correct option

- 0.6
- 0.5
- 0.4
- 0.7

Given that $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition $y = 1,\ t = 0$ Using Euler's method, y at

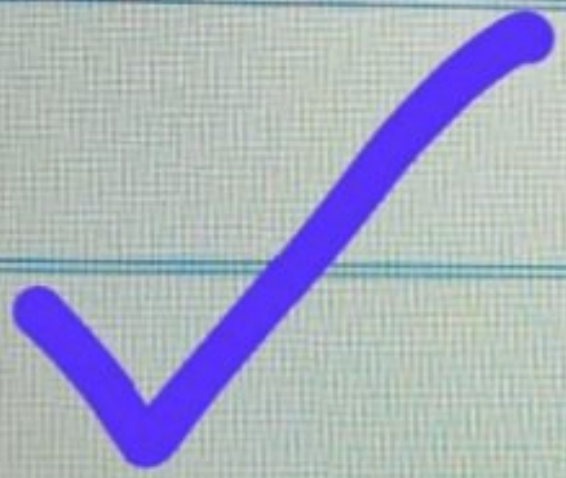
; the value of y(0.01)is

$$y_1 = y_0 + hf(t_0, y_0),$$

Select the correct option

- 1.1
- 1.2
- 1.01
- 1.02

y(0.01) = 1+0.01(1)
= 1.01



Given that $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition y=1.01 at t=0.01. Using Euler's method, y at t= 0.04, h=0.05, the value of y(0.05) is

Select the correct option

- 1.059
- 1.068
- 1.023
- 1.034

$$y_1 = y_0 + hf(t_0, y_0),$$

y(0.05) = 1.01+(0.05)(1/1.02)
= 1.01+0.049
1.059

In solving the differential equation

$$y' = x + y \; ; \; y(0) = 1$$

$$h = 0.2$$

By Euler's method y(0.2) is calculated as

Select the correct option

○	1.01
○	0.2
○	1.02
○	0.1

$$y_1 = y_0 + hf(t_0, y_0),$$

=1+0.2(1+0)

1.2

Generally, Adams methods are superior if output at many points is needed.

Select the correct option

- FALSE
- TRUE

correct

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 1$ find the 3rd term in Taylor series when t=1.5 and y/ =1 and y// =1.2

Select the correct option

Rel

○	0.1
○	0.4
○	0.3
○	0.2

correct

The predictor-corrector method an _____ method.

Select the correct option

- implicite
- indirect
- direct
- explicit

correct

We can improve the accuracy of trapezoidal and Simpson's rules using .....

Select the correct option

- Simpson's 1/3 rule
- None of the given choices
- Simpson's 3/8 rule
- Richardson's extrapolation method

Romberg's integration method.

correct

Given that $\frac{dy}{dx} = xy^{\frac{1}{3}}$ with the initial condition $y_0 = 1$ at $x_0 = 1$ and h= 0.1. Then, K_1 of Runge-Kutta method is:

$$k_1 = hf(t_n, y_n),$$

Select the correct option

- 0.4
- 0.1
- 0.2
- 0.3

correct

0.1[(1)(1)1/3]

0.1(1)

0.1

To evaluate a definite integral of tabular function f(x), piecewise linear approximation led to --------.

Select the correct option

- Simpson's 3/8 Rule
- Romberg's Method
- Simpson's 1/3 Rule
- Trapezoidal Method

correct

If $yn+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_2 is:

Select the correct option

○	$K_2 = hf(t_n + \frac{h}{2}, y_n + \frac{k_1}{2})$ correct
○	$K_2 = hf(t_n + \frac{h}{2}, y_n + \frac{k}{2})$
○	$K_2 = hf(t_n, y_n)$
○	$K_2 = hf(t_n - \frac{h}{2}, y_n - \frac{k_1}{2})$

If $ym+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_4 is:

Select the correct option

- $K_4 = hf(t_n - h, y_n - K_3)$
- $K_4 = hf(t_n, y_n)$
- $K_4 = hf(t_n + h, y_n + K_3)$ correct
- $K_4 = hf(t_n + \frac{h}{2}, y_n + \frac{k_3}{2})$

In Romberg's method, accuracy of Simpson and Trapezoidal rules is improved by --------.

Select the correct option

- extrapolation

correct

- interpolation

Geometrically the definite integral of any continuous function f(x) in the interval [a,b] gives ---------.

Select the correct option

- ○ Area of Trapezoid with dimension of 'a' and 'b'

Correct

- ○ Length of segment AB on real line
- ○ Area under f(x) on [a,b]

correct

- ○ Volume with dimensions f(x), 'a' and 'b'

Given that $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition $y = 1,\ t = 0$ find the 3rd term in Taylor series when t=0.3 and y//= 0.2.

Select the correct option

correct

- 0.018
- 0.012
- 0.014
- 0.016

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 1$ find the 2nd term in Taylor series when t=1.5 and y/ =1.2

Select the correct option

○	0.7
○	0.4
○	0.5
○	0.6

correct

Given that $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition $y = 1,\ t = 0$ Using Euler's method, y at

$$h = 0.01$$

; the value of y(0.01)is

Select the correct option

○	1.1
○	1.01
○	1.02
○	1.2

correct

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 1$ find the 3rd term in Taylor series when t=1.5 and y/ =1 and y// =1.2

Select the correct option

Reload M

O	0.1
O	0.3
O	0.2
O	0.4

Question # 3 of 10 (Start time: 08:30:11 PM, 09 March 2022)

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 3rd term in Taylor series when t=1.5 and y// =0.6

Select the correct option

○	1.45
○	1.25
○	1.15
○	1.35

correct

Question # 4 of 10 (Start time: 08:31:02 PM, 06 March 2022)

Adam Moulton is also known as _____ method.

Select the correct option

- linear multistep
- non linear single step
- non linear multistep
- linear single step

correct

If $\frac{dy}{dx} = \frac{y+x}{y-x}, y_0 = 1$ at $x_0 = 0$ and h= 0.2. Then, K_1 of Runge-Kutta method is:

Select the correct option

- 0.4
- 0.2
- 0.1
- 0.3

$$k_1 = hf(t_n, y_n)$$

correct

In Simpson's rule, we assume that f(x) is continuous on [a, b] and we divide [a, b] into an ..... number n of subintervals of equal length.

Select the correct option

○	Prime
○	None of the given choices
○	Even
○	Odd

correct

Given that $\frac{dy}{dt} = 1 - \sqrt{y}$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 2nd term of Taylor series when t=0.5 and y/ =0.

Select the correct option

○	0.1
○	0 correct
○	0.2
○	0.3

Trapezoidal and Simpson's integrations are just a linear combination of values of the
given function at different values of the .....variable.

Select the correct option

- None of the given choices
- Independent

correct

- Dependent
- Arbitrary

If $yn+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_4 is:

Select the correct option

○	$K_4 = hf(t_n, y_n)$
○	$K_4 = hf(t_n + h, y_n + K_3)$ correct
○	$K_4 = hf(t_n - h, y_n - K_3)$
○	$K_4 = hf(t_n + \frac{h}{2}, y_n + \frac{k_3}{2})$

Given that $\frac{dy}{dt} = 1 - \sqrt{y}$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 3rd term of Taylor series when t=0.5 and y// =0.25.

Select the correct option

Reload M

○	0.01425
○	0.01625
○	0.01325
○	0.01525

correct

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 3rd term in Taylor series when t=1.5 and $y''' = 0.6$

Select the correct option

- 1.15
- 1.45
- 1.35
- 1.25

Given that $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition $y = 1, \ t = 0$ find the 2nd term in Taylor series when t=0.2 and y/= 0.3.

Select the correct option

- 0.06
- 0.01
- 0
- 0.02

Given that $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition $y = 1,\ t = 0$ find the 3rd term in Taylor series when t=0.3 and y//= 0.2.

Select the correct option

○	0.018
○	0.016
○	0.012
○	0.014

If $y_{n+1} = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_4 is:

Select the correct option

- $K_4 = hf(t_n + \frac{h}{2}, y_n + \frac{k_3}{2})$
- $K_4 = hf(t_n - h, y_n - K_3)$
- $K_4 = hf(t_n, y_n)$
- $K_4 = hf(t_n + h, y_n + K_3)$ ✓

correct

A desirable feature of a multistep method is that the loca truncation error (L. T. E.) can be determined and a correction term can be included, which improves the accuracy of the answer at each step.

Select the correct option

- TRUE

correct

- FALSE

If $\frac{dy}{dx} = \frac{y+x}{y-x}, y_0 = 1$ at $x_0 = 0$ and h= 0.2. Then, K_2 of Runge-Kutta method is:

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

Select the correct option

- 0.22
- 0.42
- 0.32
- 0.12

correct

Multistep method does not improves the accuracy of the answer at each step.

elect the correct option

- FALSE
- TRUE

correct

Adam Moulton is also known as _____ method.

Select the correct option

- linear multistep
- non linear multistep
- linear single step
- non linear single step

correct

Given that $\frac{dy}{dt} = t + \backslash y$ with the initial condition $y_0 = 1$ at $t_0 = 0$ and h= o.1. Then, K_1 of Rung

elect the correct option

- 0.5
- 0.2
- 0.1
- 0.4

correct

The area of a trapezoid is obtained by adding the area of a .....and a triangle.

Select the correct option

- Square
- None of the given choices
- Circle
- Rectangle

correct

In double integration, we keep one variable say x fixed and .....

Select the correct option

- Reliable the other variable y
- Varying the other variable y

correct

The predictor-corrector method an _____ method.

lect the correct option

- indirect
- implicite
- explicit
- direct

correct

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 3rd term in Taylor series when t=1.5 and y// =0.6

Select the correct option

○	1.15
○	1.25
○	1.45
●	1.35

correct

If $\frac{dy}{dx} = \frac{y+x}{y-x}$, $y_0 = 1$ at $x_0 = 0$ and h= 0.2. Then, K_2 of Runge-Kutta method is:

Select the correct option

- 0.42
- 0.12
- 0.32 (correct)
- 0.22

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

Given that $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition y=1.01 at t=0.01. Using Euler's method, y at t= 0.04, h=0.05, the value of y(0.05) is

Select the correct option

- 1.023
- 1.034
- 1.068
- 1.059

correct

If $yn+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_3 is:

Select the correct option

○	$K_3 = hf(t_n + \frac{h}{2}, y_n + \frac{k_1}{2})$
○	$K_3 = hf(t_n, y_n)$
○	$K_3 = hf(t_n - \frac{h}{2}, y_n - \frac{k_2}{2})$
●	$K_3 = hf(t_n + \frac{h}{2}, y_n + \frac{k_2}{2})$ correct

Question # 9 of 10 (Start time: 09:32:56 PM, 06 March 2022)

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 2nd term in Taylor series when t=1, y/ =1.2, and h=0.1.

Select the correct option

Rel

- 1.3
- 1.4
- 1.2 ●
- 1.5

correct

Click to Save Answer & Move to Ne

Geometrically the definite integral of any continuous function f(x) in the interval [a,b] gives ---------.

elect the correct option

- ○ Volume with dimensions f(x), 'a' and 'b'
- ● Area under f(x) on [a,b]

correct

- ○ Area of Trapezoid with dimension of 'a' and 'b'

Correct

- ○ Length of segment AB on real line

Which of the following method is simplest one to integrat
numerically a given tabular function but give more error?

Select the correct option

- Simpson's 3/8 Rule
- Rectangular method
- Trapezoidal method
- Simpson's 1/3 Rule



Correct

To evaluate a definite integral of tabular function f(x), piecewise quardratic approximation led to ———

Select the correct option

- Trapezoidal Method
- Simpson's Rule

correct

We can improve the accuracy of trapezoidal and Simpson's rules using .....

Select the correct option

- Simpson's 3/8 rule
- None of the given choices
- Richardson's extrapolation method
- Simpson's 1/3 rule

correct

To evaluate a definite integral of tabular function f(x), piecewise quardratic approximation led to -------

Select the correct option

- Simpson's Rule

correct

- Trapezoidal Method

A desirable feature of a multistep method is that the local truncation error (L. T. E.) can be determined and a correction term can be included, which improves the accuracy of the answer at each step.

Select the correct option

- FALSE
- TRUE

correct

Adams – Bashforth is a multistep method.

Select the correct option

- FALSE
- TRUE

correct

Given that $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition $y = 1,\ t = 0$ find the 2nd term in Taylor series when t=0.2 and y/= 0.3.

Select the correct option

○	0.02
○	0.06
○	0
○	0.01

correct

Question # 9 of 10 (Start time: 08:45:43 PM, 06 March 2022) Total Marks: 1

Which of the following reason(s) lead towards the numerical integration methods?

Select the correct option

- All above choices are true

Correct

- Integrand is given in tabular form
- Analytical evaluation of integral is impossible
- Analytical evaluation of integral is very complicated

correct

Generally, Adams methods are superior if output at _____ points is needed.

Select the correct option

○	at most
○	two
●	many
○	single

Given that $\frac{dy}{dt} = t + \sqrt{y}$ with the initial condition $y_0 = 1$ at $t_0 = 0$ Using Modified Euler's method, for the range $0 \leq t \leq 0.6$, h = 0.1 is

$$y_1^{(1)} = y_0 + hf(t_0, y_0)$$

Select the correct option

Re

○	1.1
○	1.0
●	2.1
○	1.2

1+0.1(0+1)

= 1.1

$$y_1 = y_0 + h\frac{f(t_0, y_0) + f(t_1, y_1^{(1)})}{2}$$

1+0.2[1+(0.1+(1.1)sqrt

1+0.2[1+1.14]

1+0.2(2.14)/2

1+

If $ym + 1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_4 is:

Select the correct option

- $K_4 = hf(t_n + \frac{h}{2}, y_n + \frac{k_3}{2})$
- $K_4 = hf(t_n - h, y_n - K_3)$
- $K_4 = hf(t_n, y_n)$
- $K_4 = hf(t_n + h, y_n + K_3)$

Simpson's rule is a numerical method that approximates the value of a definite integral by using ..... polynomials.



Select the correct option

- Linear
- None of the given choices
- Quadratic

Correct

- Cubic

Question # 1 of 10 (start time: 07:13:20 PM, 08 September 2022)

Given that $\frac{dy}{dt} = t + y$ with the initial condition: $y_0 = 1$ at $t_0 = 1$ and h= 0.1 Then, K_1 of Runge-Kutta method is:

Select the correct option

Re

○	0.2
○	0.4
○	0.3
○	0.1

Correct

Question # 2 of 10 (start time: 07:14:16 PM, 08 september 2022)

If $\frac{dy}{dx} = \frac{y+x}{y-x}$, $y_0 = 1$ at $x_0 = 0$ and h= 0.2 Then, K_2 of Runge-Kutta method is:

Select the correct option

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

- ○ 0.22
- ○ 0.32
- ○ 0.12
- ○ 0.42

Correct

Question # 3 of 10 (Start time: 07:14:59 PM, 08 September 2022) **Total Marks:** 1

Given that $\frac{dy}{dt} = t + \sqrt{y}$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 3rd term in Taylor series when t=1, y/ =0.2, y// =2, and h=0.1

Select the correct option

Reload Math Equations

○	0.1
○	2
○	0
○	1

Question # 4 of 10 (Start time: 07:16:06 PM, 08 September 2022)

Given that $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition $y = 1,\ t = 0$ find the 3rd term in Taylor series when t=0.3 and y//= 0.2.

Select the correct option

Reloc

○	0.018
○	0.014
○	0.012
○	0.016

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 3rd term in Taylor series when t=1.5 and y// =0.6

Select the correct option

Reload Math

○	1.25
○	1.35
○	1.45
○	1.15

Question # 8 of 10 (start time: 07:19:19 PM, 08 september 2022)

If $yn+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_3 is:

Select the correct option

○	$K_3 = hf(t_n + \frac{h}{2}, y_n + \frac{k_1}{2})$
○	$K_3 = hf(t_n + \frac{h}{2}, y_n + \frac{k_2}{2})$ Correct
○	$K_3 = hf(t_n, y_n)$
○	$K_3 = hf(t_n - \frac{h}{2}, y_n - \frac{k_2}{2})$

Question # 9 of 10 (Start time: 07:20:05 PM, 08 September 2

Adam Moulton is also known as _____ method.

Select the correct option

- ○ non linear single step
- ○ linear multistep
- ○ linear single step

C

- ○ non linear multistep

Question # 10 of 10 (Start time: 05:12:19 PM, 08 September 2022)

If $yn+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_2 is:

Select the correct option

○	$K_2 = hf(t_n - \frac{h}{2}, y_n - \frac{k_1}{2})$
○	$K_2 = hf(t_n + \frac{h}{2}, y_n + \frac{k}{2})$
○	$K_2 = hf(t_n + \frac{h}{2}, y_n + \frac{k_1}{2})$
○	$K_2 = hf(t_n, y_n)$

CO

Question # 6 of 10 (Start time: 05:10:06 PM, 08 September 2022) **Total Marks: 1**

In Simpson's Rule, we use parabolas to approximate each part of the curve. This proves to be very efficient as compared to Trapezoidal rule.

Select the correct option

- FALSE
- TRUE

Correct

If $yn+1 = y_n + \frac{1}{6}(K_1 + 2K_2 + 2k_3 + k_4)$ then, K_4 is:

Select the correct option

○	$K_4 = hf(t_n, y_n)$
○	$K_4 = hf(t_n + h, y_n + K_3)$ Correct
○	$K_4 = hf(t_n + \frac{h}{2}, y_n + \frac{k_3}{2})$
○	$K_4 = hf(t_n - h, y_n - K_3)$

In Simpson's Rule, we use parabolas to approximate each part of the curve. This proves to be very efficient as compared to Trapezoidal rule.

Select the *correct* option

- ○ FALSE
- ○ TRUE

Correct

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 1$ find the 3rd term in Taylor series when t=1.5 and y/ =1 and y// =1.2

Select the correct option

Reload Math Equations

○	0.3
○	0.1
○	0.4
○	0.2

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 1$ find the 2nd term in Taylor series when $h=0.5$ and $y' = 0.2$

Select the correct option

Reload M

- 0.7
- 0.5
- 0.6
- 0.4

Click to Save Answer & Move to N

Given that $\frac{dy}{dt} = 1 - \sqrt{y}$ with the initial condition $y_0 = 1$ at $t_0 = 0$ find the 3rd term of Taylor series when t=0.5 and y// =0.25.

select the correct option

Reload Math E

○	0.01625
○	0.01325
○	0.01525
○	0.01425

The second langrange polynomial with equally spaced nodes produced the formula for _____

Select the correct option

- ○ Richardson's method
- ○ newton's method
- ○ Simpson's rule
- ○ Trapezoidal rule

Question # 8 of 10 (Start time: 05:10:15 PM, 08 September 2022)

Given that $\frac{dy}{dt} = t + y$ with the initial condition $y_0 = 1$ at $t_0 = 1$ and h= 0.1 Then, K_1 of Runge-Kutta method is:

Select the correct option

○	0.3
○	0.1
○	0.2
○	0.4

correct

Question # 6 of 10 (Start time: 05:10:06 PM, 08 September 2022) **Total Marks: 1**

In Simpson's Rule, we use parabolas to approximate each part of the curve. This proves to be very efficient as compared to Trapezoidal rule.

Select the correct option

- FALSE
- TRUE

Multistep method does not improves the accuracy of the answer at each step.

Select the correct option

- ○ TRUE
- ○ FALSE

Question # 5 of 10 (Start time: 05:09:26 PM, 08 September 2022)

Given that $\frac{dy}{dx} = xy^{\frac{1}{3}}$ with the initial condition $y_0 = 1$ at $x_0 = 1$ and h= 0.1. Then, K_1 of Runge-Kutta method is

Select the correct option

- ○ 0.3
- ○ 0.2
- ○ 0.1
- ○ 0.4

Correct

Question # 6 of 10 (Start time: 05:09:06 PM, 08 September 2022)

If $\frac{dy}{dx} = \frac{y+x}{y-x}$, $y_0 = 1$ at $x_0 = 0$ and h= 0.2 then, K_1 of Runge-Kutta method is:

Select the correct option

- 0.4
- 0.3
- 0.1
- 0.2

Question # 4 of 10 (Start time: 05:08:46 PM, 08 September 2022)

Euler's Method numerically computes the approximate _____ of a function.

Select the correct option

- ○ derivative

Correct

- ○ error
- ○ antiderivative
- ○ value

Question # 5 of 10 (Start time: 05:08:31 PM, 08 September 2022) **Total Marks:** 1

A desirable feature of a multistep method is that the local truncation error (L. T. E.) can be determined and a correction term can be included, which improves the accuracy of the answer at each step.

Select the correct option

○	TRUE
○	FALSE

Question # 2 of 10 (Start time: 05:05:43 PM, 08 September 2022) **Total Marks:** 1

The need of numerical integration arises for evaluating the definite integral of a function that has no explicit _____ or whose antiderivative is not easy to obtain.

Select the correct option

- ○ derivatives

Correct

- ○ antiderivative

Euler's Method numerically computes the approximate derivative of a function.

Select the correct option

- ○ FALSE
- ○ TRUE

Correct

Question # 2 of 10 (Start time: 05:07:00 PM, 08 September 2022)

If $\frac{dy}{dx} = \frac{y+x}{y-x}$, $y_0 = 1$ at $x_0 = 0$ and h= 0.2 then, K_2 of Runge-Kutta method is:

Select the correct option

- 0.42
- 0.12
- 0.32
- 0.22

Correct